

DESIGN OF EMBEDDED SYSTEMS AND IOT

Team 2

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1. OVERVIEW

PROBLEM

- Context: Rapid urbanization has caused severe traffic congestion.
- Challenge: Inefficient management forces drivers to spend excessive time searching for parking spots.
- Impact: Leads to fuel wastage and increased environmental emissions.

SOLUTION

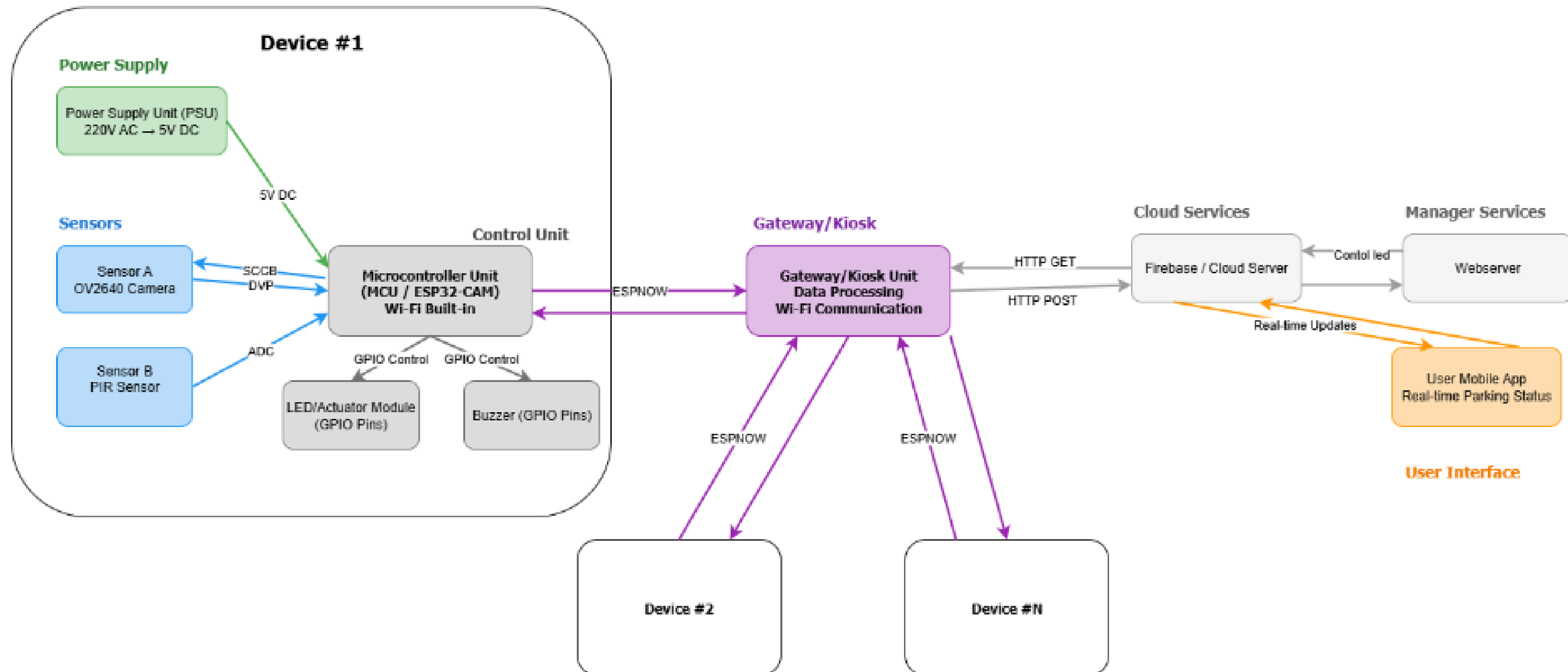
Embedded IoT-based Smart Parking System:

- Facilitate real-time monitoring and management.
- Optimize space utilization to enhance user experience.

2. SYSTEM DESIGN

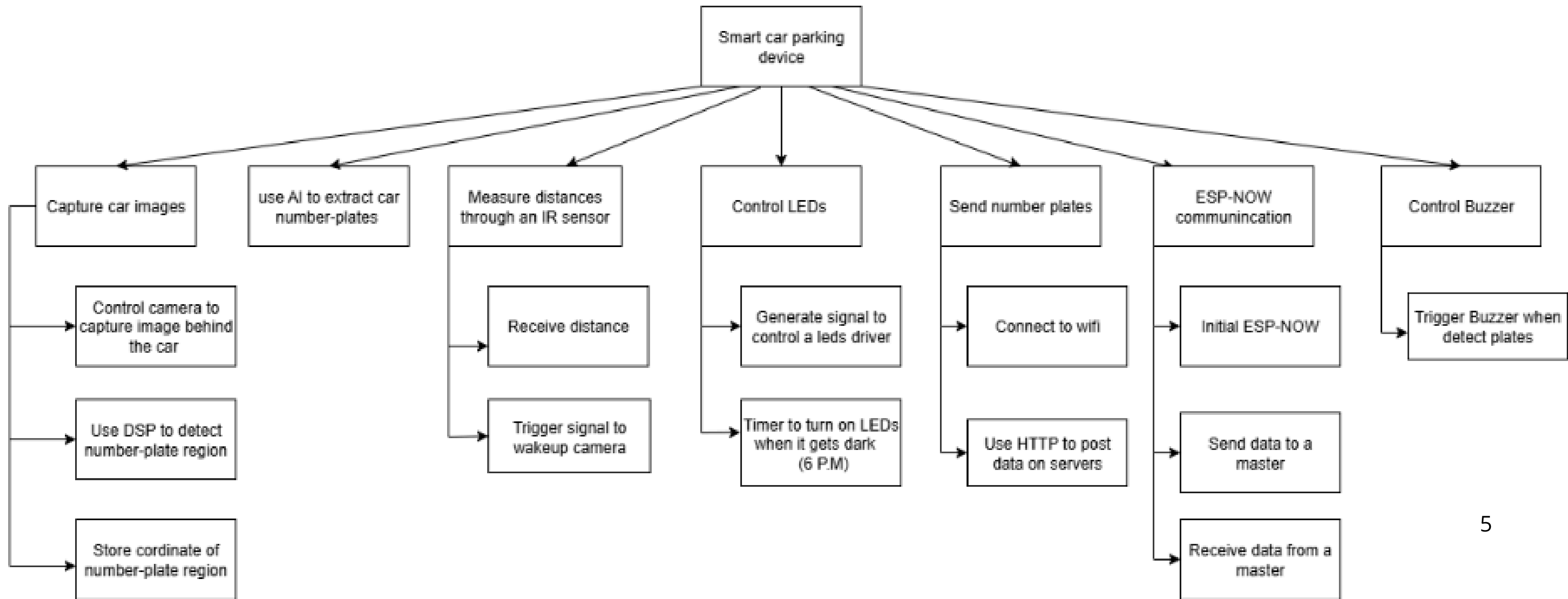
Hardware Block Diagram

Smart Parking System Block Diagram



2. SYSTEM DESIGN

Functional decomposition:



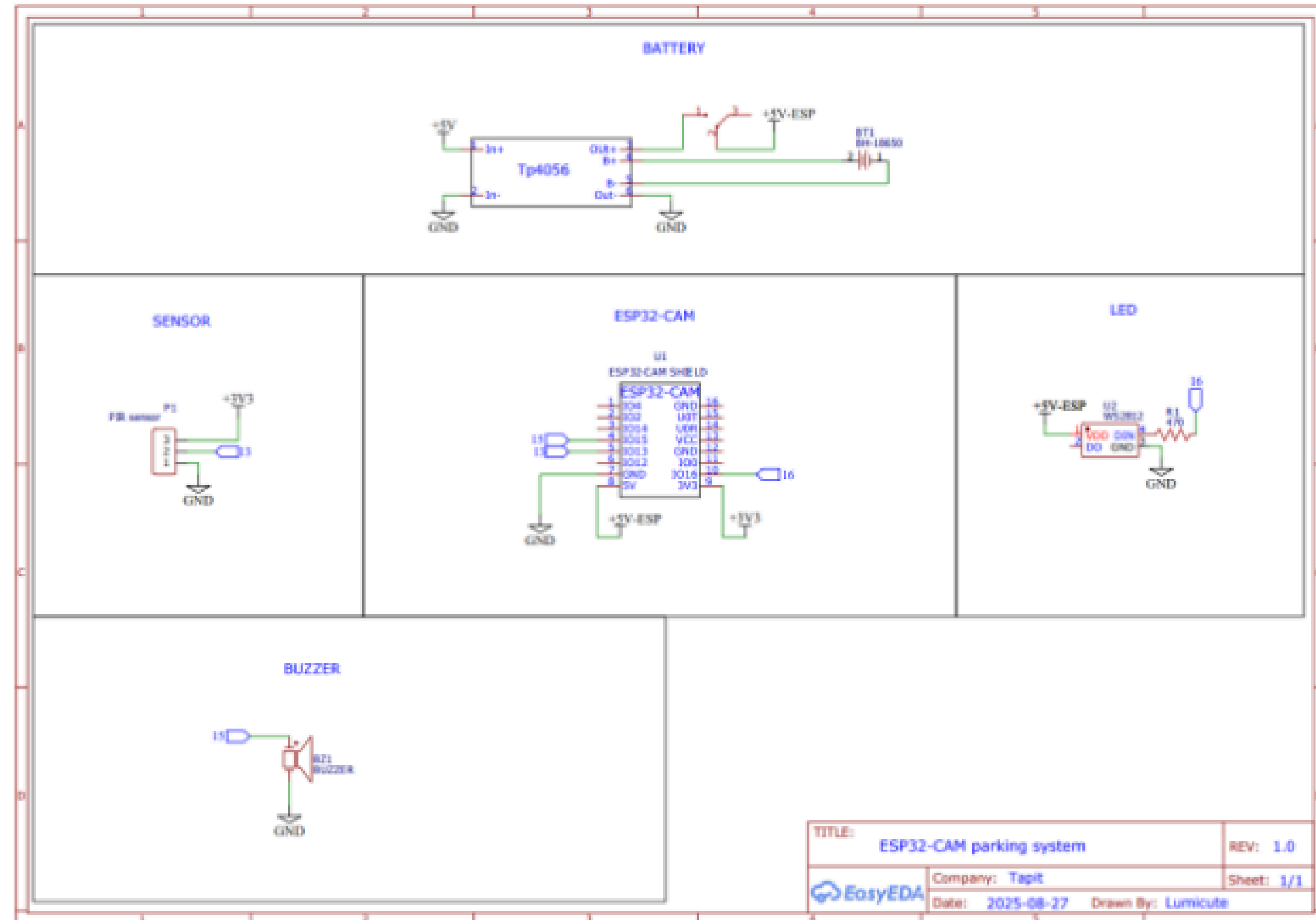
2. SYSTEM DESIGN

Bill of Materials

No.	Component Name	Quantity	Price (đồng)	Total Price (đồng)	Note
1	Adapter	1	40	40	Available
2	TP 4056	1	5	5	Available
3	18650 battery	1	20	20	Available
4	ESP 32 CAM OV 2640	1	180	180	Available
5	PIR Sensor	1	60	60	
6	ESP 32	1	79	79	Available
7	LED RGB	1	5	5	
8	Case	1	50	50	Available
Total				439	

2. SYSTEM DESIGN

Schematic diagram



2. SYSTEM DESIGN

Software layer

Software layer for node

Application layer

GPIO: gpio_set_direction(), gpio_set_level(), gpio_get_level()

ADC: adc1_get_raw(), adc2_get_raw()

Timer: TaskDelay(), esp_timer_get_time().

PWM: ledc_set_duty(), ledc_update_duty().

Middleware Layer

Wi-Fi: wifi_init_sta(), esp_wifi_connect()

HTTP: esp_http_client_perform()

Camera: esp_camera_init(), esp_camera_fb_get()

DSP: dsp_image_gray(), dsp_image_resize()

AI: esp_dl_model_infer() hoặc MicroInterpreter::Invoke()

JSON: cJSON_CreateObject(), cJSON_AddStringToObject()

Espnow: esp_now_*, Wi-Fi radio/channel setup to support ESP-NOW

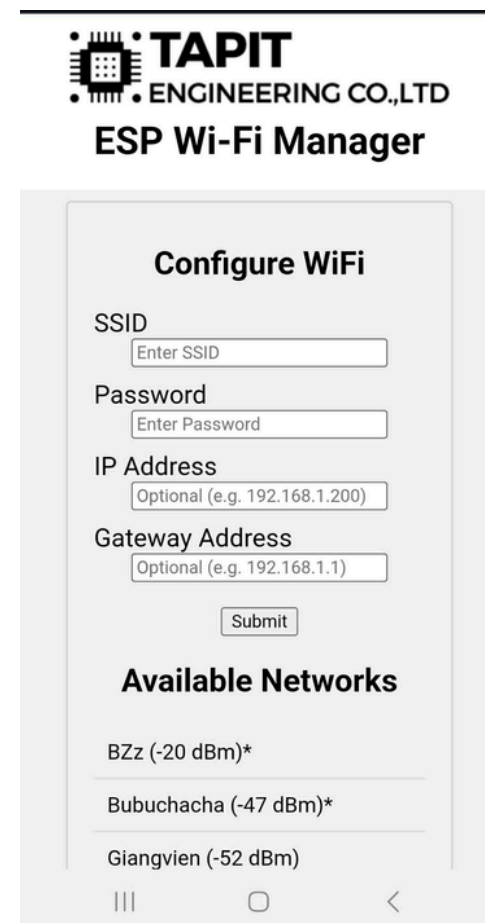
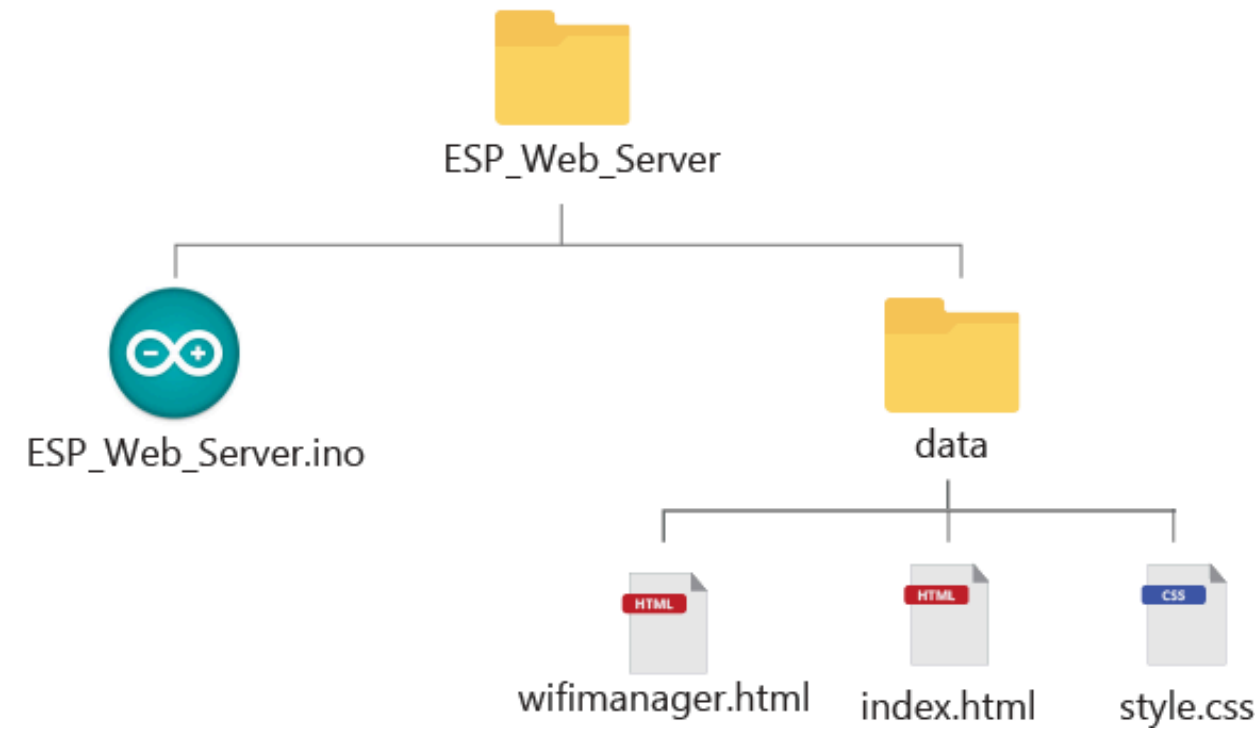
Drive layer

Camera driver: configure camera, and capture images

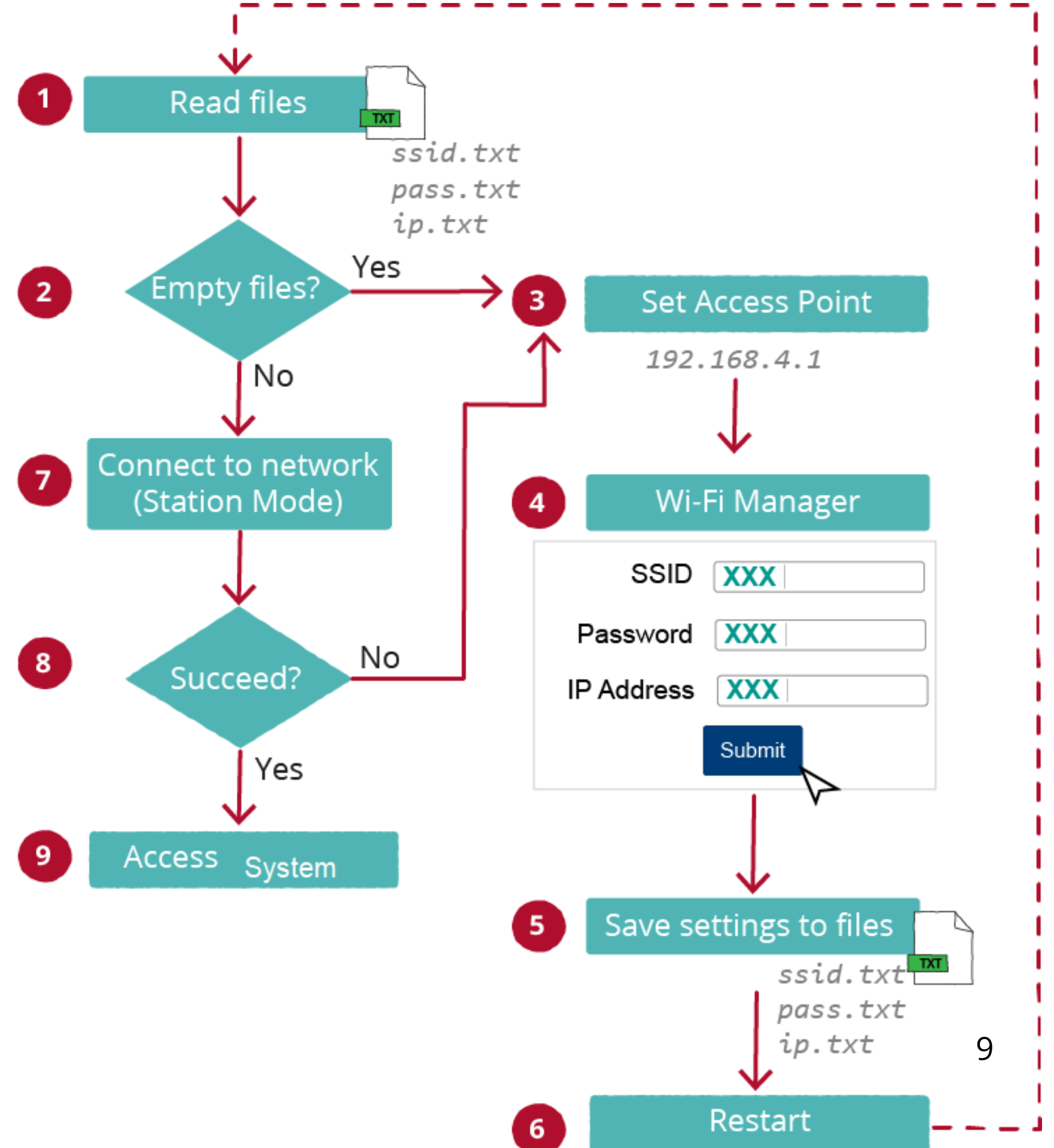
RAM driver: initialize PSRAM and allocate memory.

3. COMMUNICATION & ALGORITHMS

Wi-Fi Manager



The image shows a mobile web interface for 'TAPIT ENGINEERING CO.,LTD ESP Wi-Fi Manager'. It features a 'Configure WiFi' section with input fields for SSID, Password, IP Address (Optional), and Gateway Address (Optional), followed by a 'Submit' button. Below this is an 'Available Networks' section listing three networks: BZz (-20 dBm)*, Bubuchacha (-47 dBm)*, and Giangvien (-52 dBm).



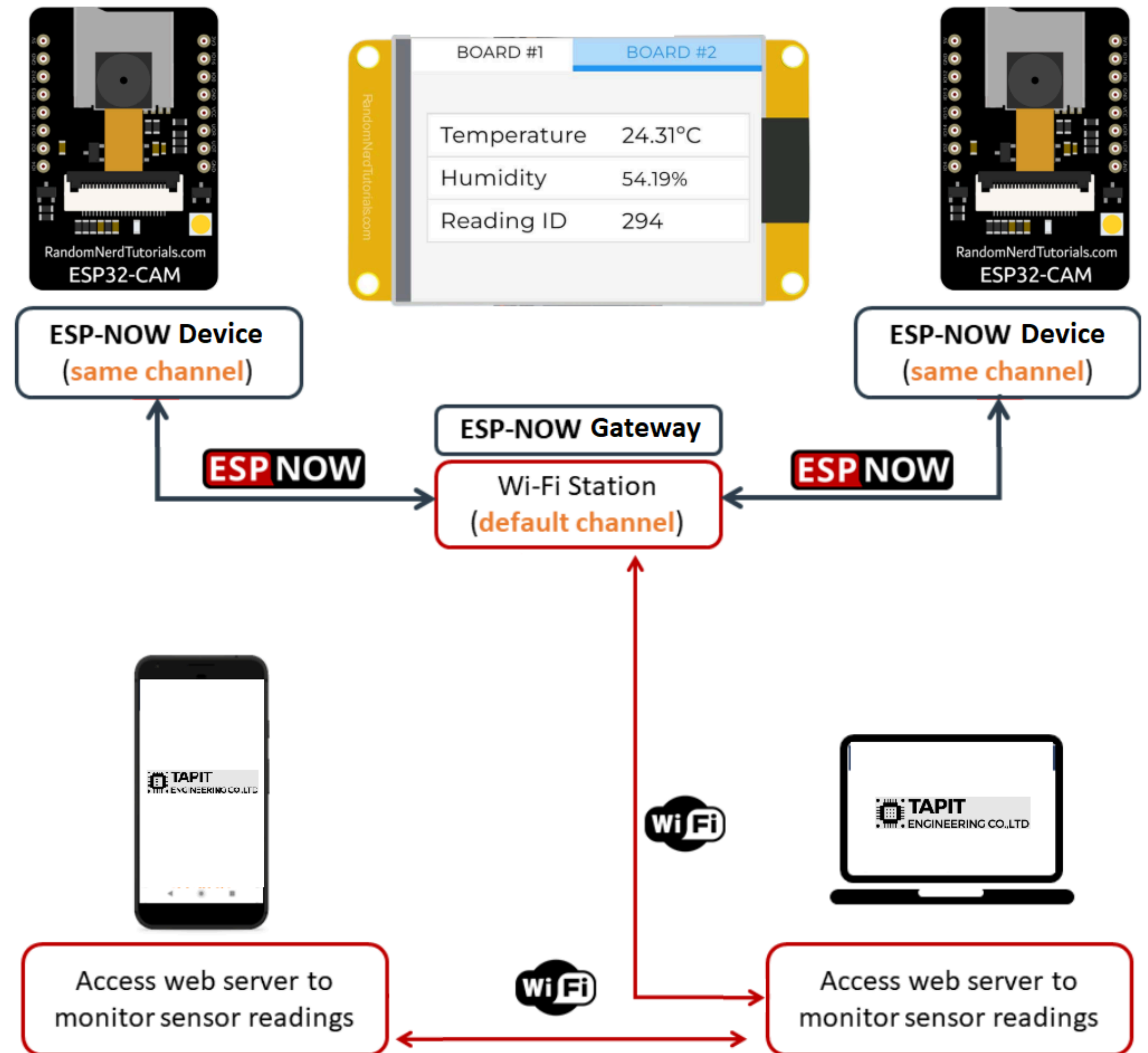


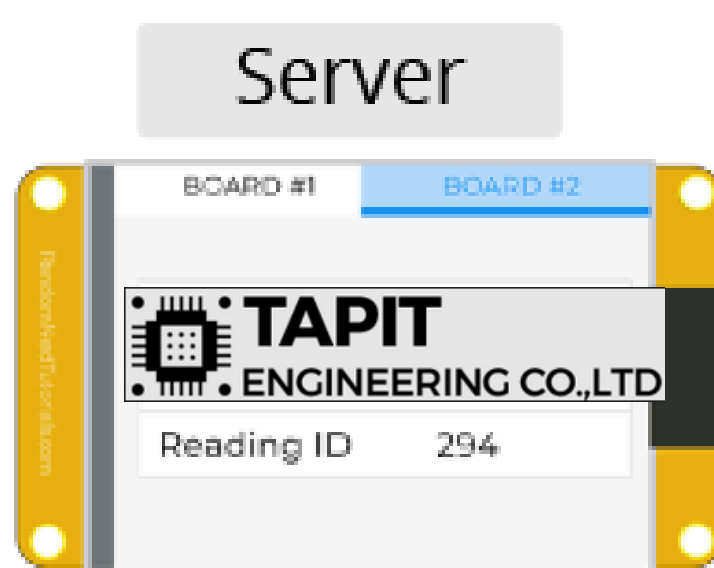
ESP-NOW is a fast, low-power wireless communication protocol developed by Espressif for ESP32 and ESP8266 microcontrollers. It enables devices to communicate directly with each other (peer-to-peer) without needing a Wi-Fi router or an internet connection.

- **Direct Connection**
- **Speed & Efficiency**
- **Range:** (up to 200+ meters in open areas)
- **Payload Limit:** Up to 250 bytes of data
- **Versatility:** one-way and two-way communication setups.



- The ESP32 Device boards must use the same Wi-Fi channel of the Gateway board.
- The Wi-Fi channel of the receiver board is automatically assigned by your Wi-Fi router.
- The Wi-Fi mode of the gateway board must be access point and station (WIFI_AP_STA).

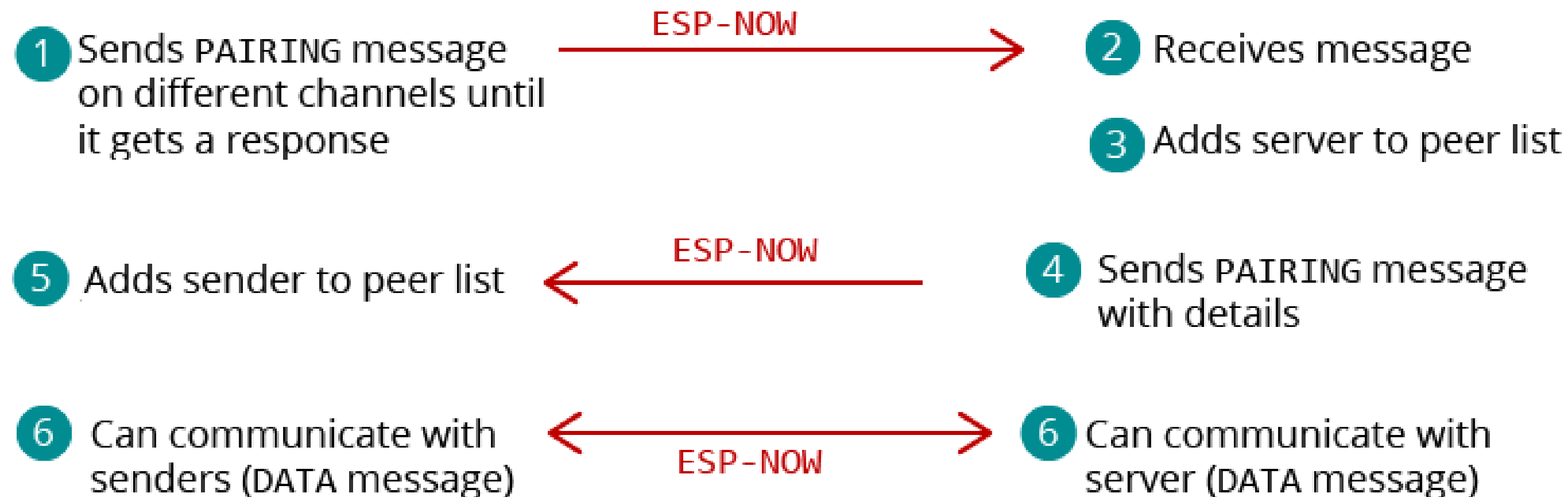
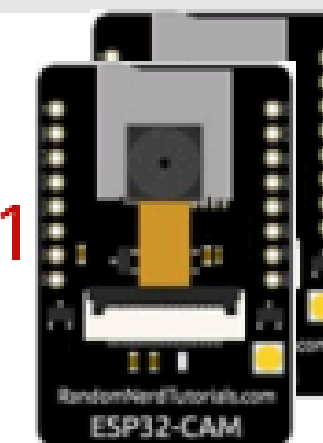


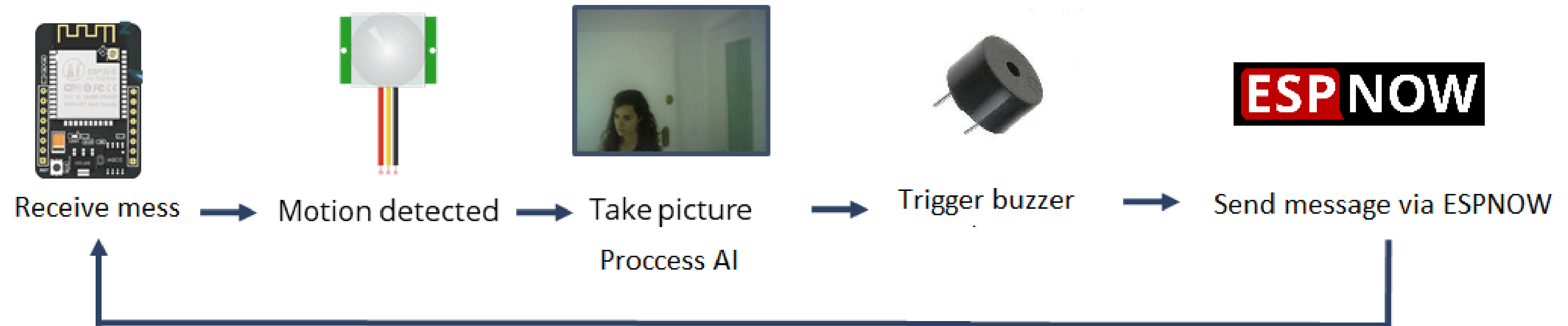


Senders

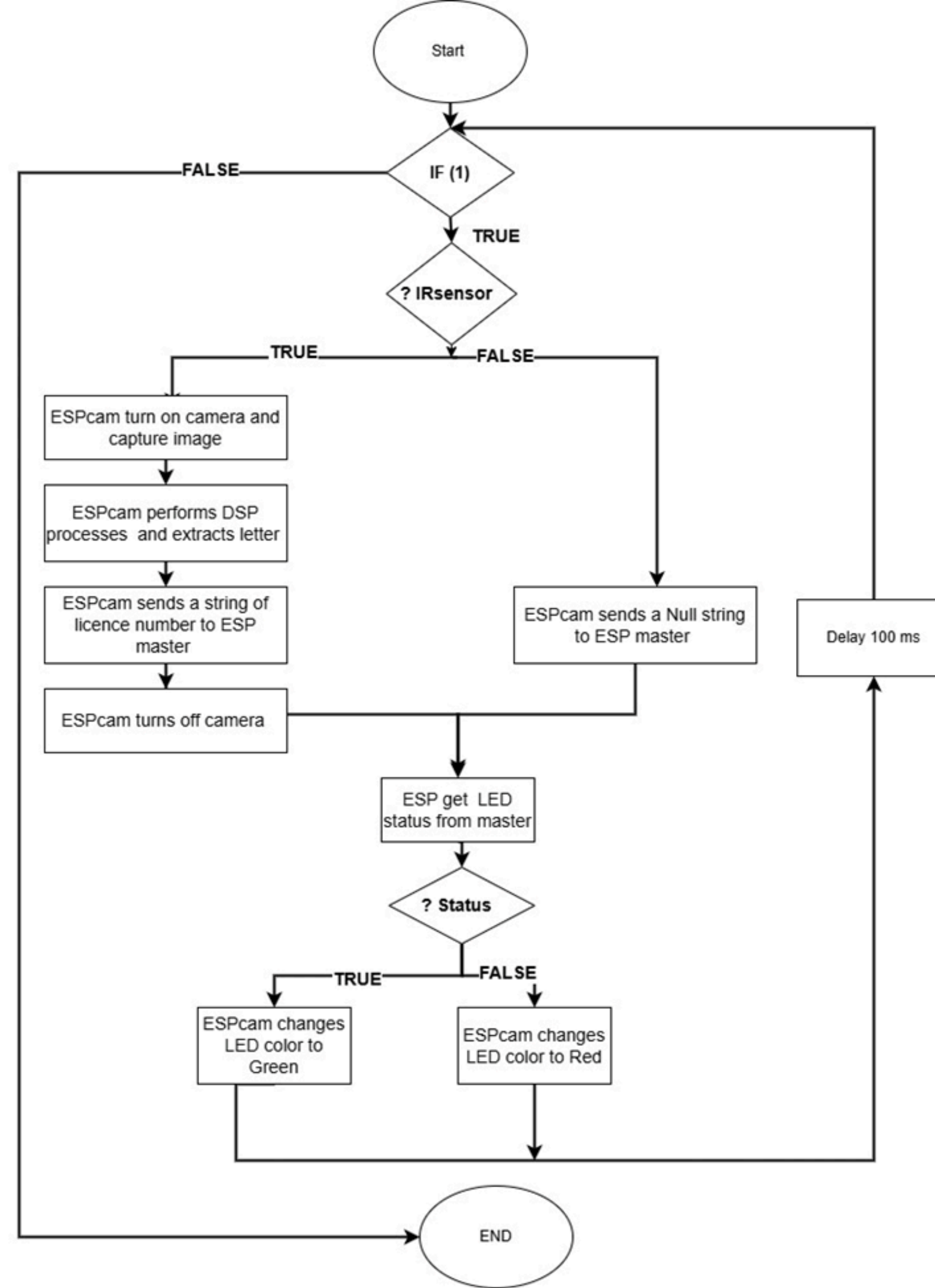
id=1

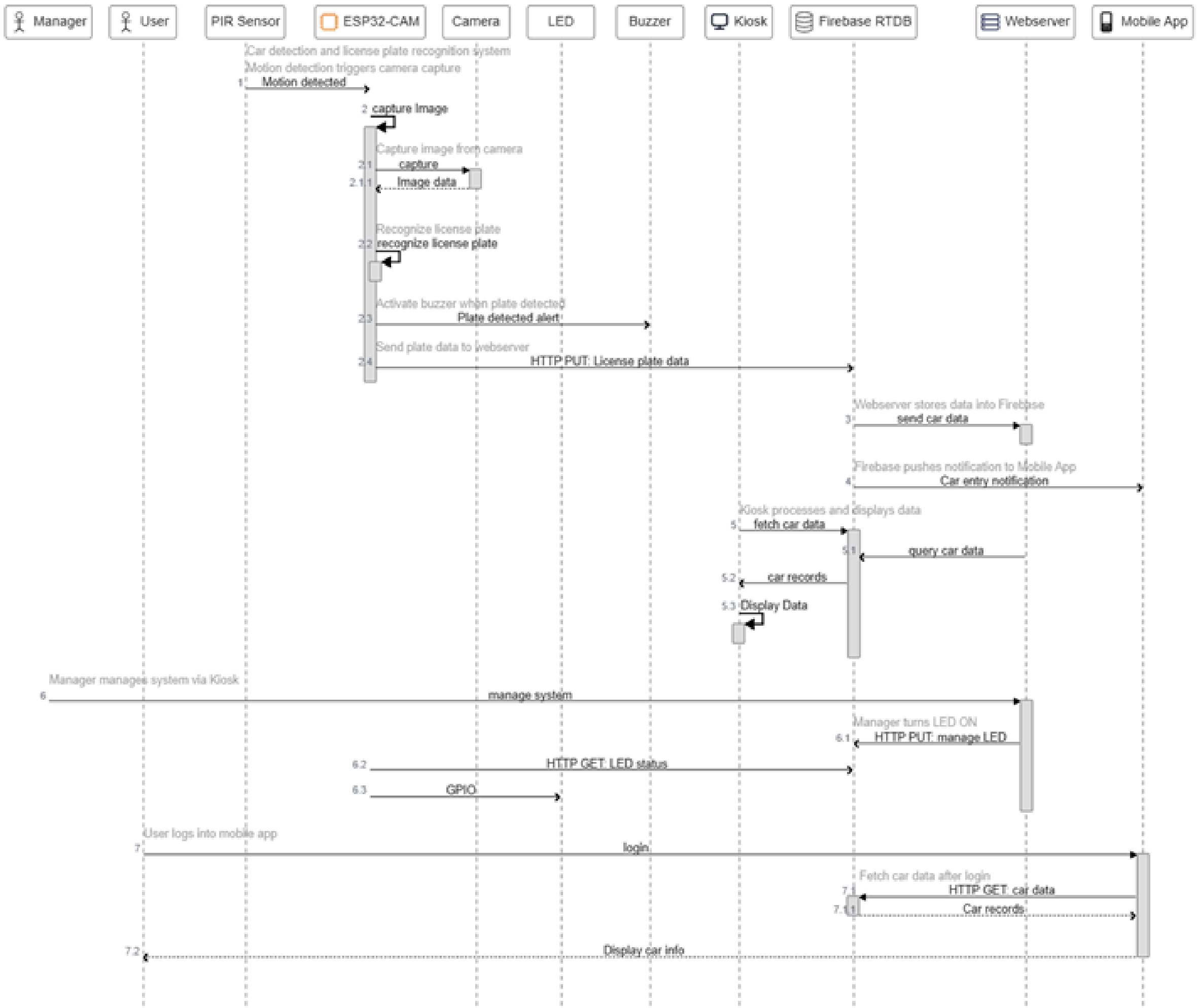
id=2





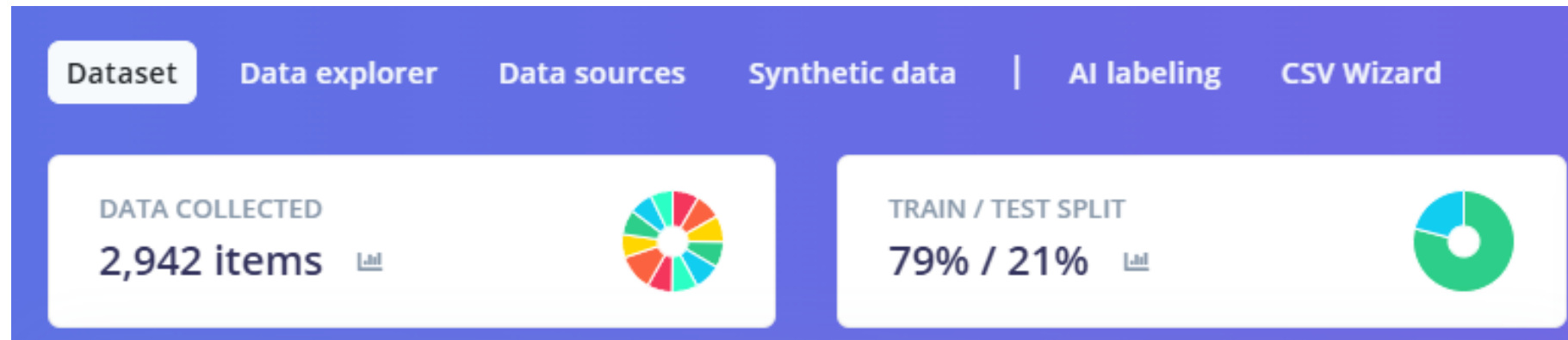
FLOWCHART



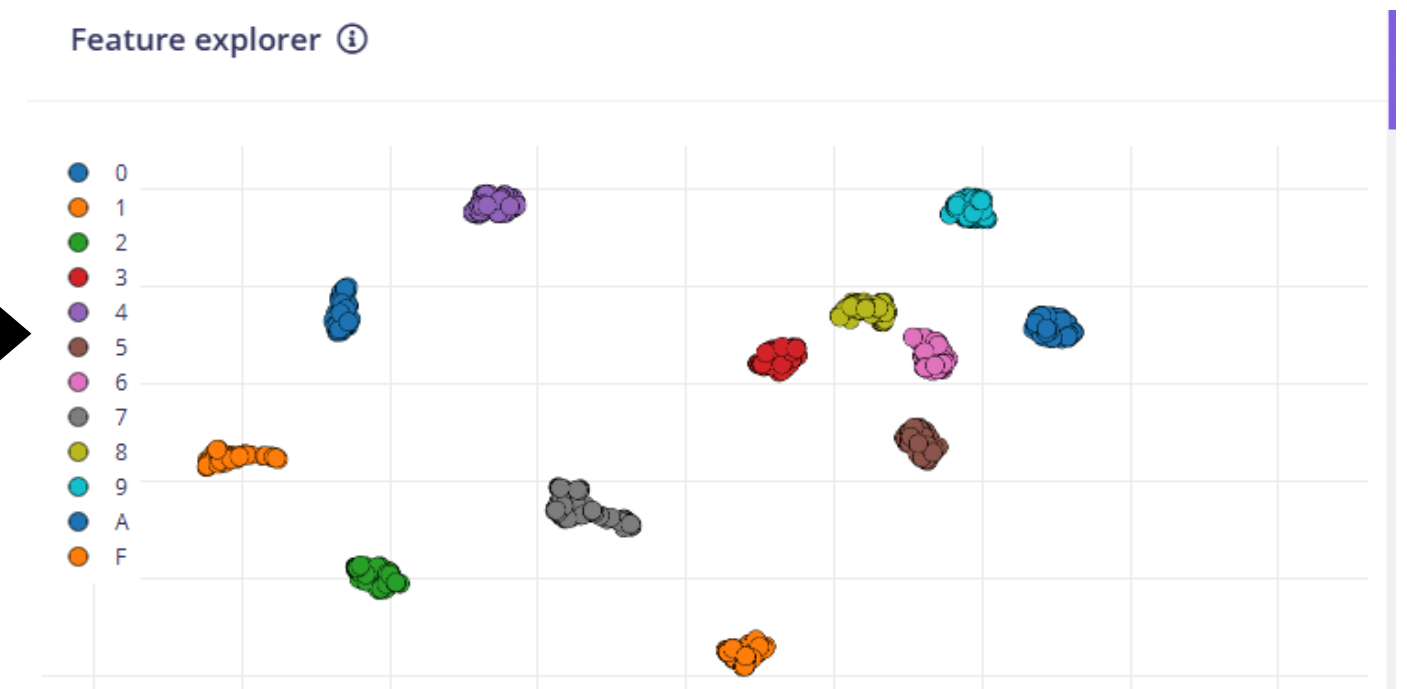
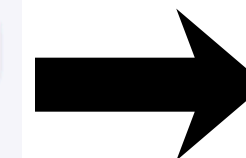
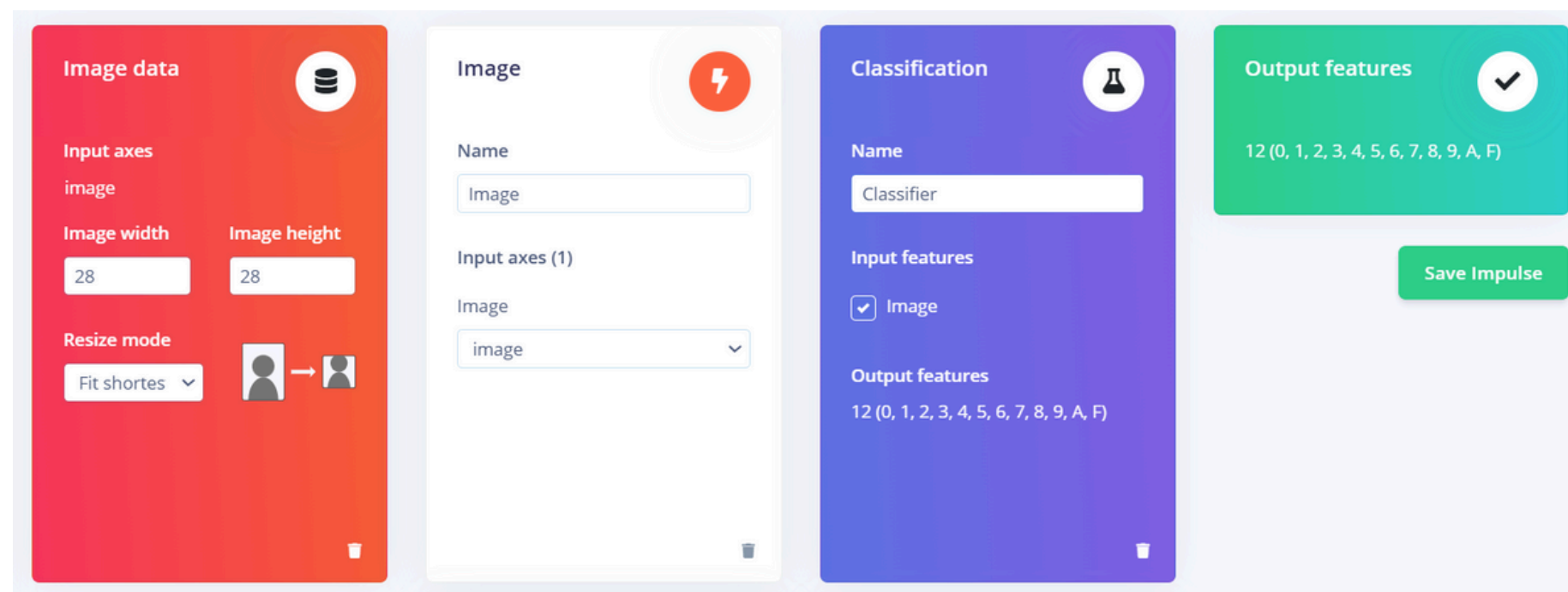


SEQUENCE DIAGRAM

4. AI MODEL TRAINING



Feature extraction of 12 characters from 0 to 9, A and F



4. AI MODEL TRAINING

%

ACCURACY
99.49%

Metrics for Classifier

Download icon

METRIC	VALUE
Area under ROC Curve ?	1.00
Weighted average Precision ?	1.00
Weighted average Recall ?	1.00
Weighted average F1 score ?	1.00

Confusion matrix

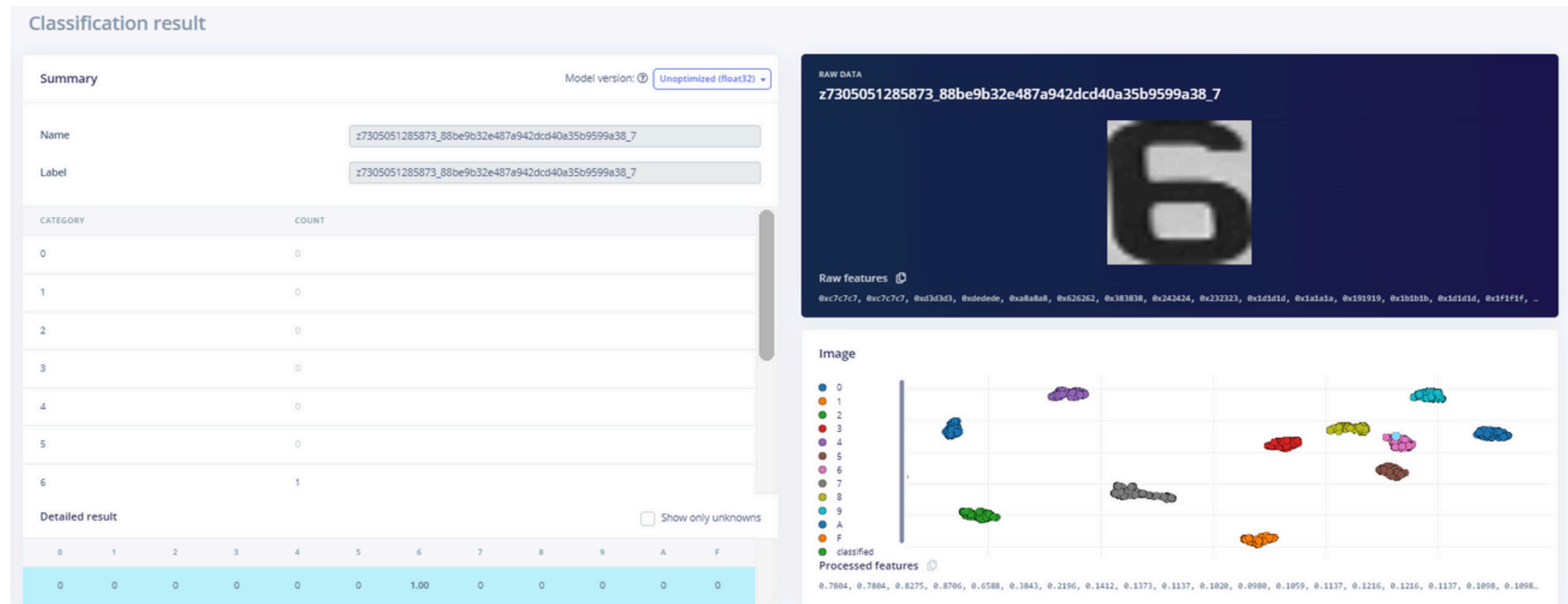
	0	1	2	3	4	5	6	7	8	9	A	F	UNCE
0	100%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
1	0%	100%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
2	0%	0%	100%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
3	0%	0%	0%	100%	0%	0%	0%	0%	0%	0%	0%	0%	0%
4	0%	0%	0%	0%	100%	0%	0%	0%	0%	0%	0%	0%	0%
5	0%	0%	0%	0%	0%	100%	0%	0%	0%	0%	0%	0%	0%
6	0%	0%	0%	0%	0%	0%	100%	0%	0%	0%	0%	0%	0%
7	0%	0%	0%	0%	0%	0%	0%	100%	0%	0%	0%	0%	0%
8	0%	0%	0%	0%	0%	0%	4.3%	0%	93.5%	0%	0%	0%	2.2%
9	0%	0%	0%	0%	0%	0%	0%	0%	0%	100%	0%	0%	0%
A	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	100%	0%	0%
F	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	100%	0%
F1 SCO	1.00	1.00	1.00	1.00	1.00	1.00	0.98	1.00	0.97	1.00	1.00	1.00	

Quantized
(int8)

Selected ✓

	IMAGE	CLASSIFIER	TOTAL
LATENCY	7 ms.	47 ms.	54 ms.
RAM	4.0K	18.0K	18.0K
FLASH	-	52.5K	-
ACCURACY			99.49%

4. AI MODEL TRAINING

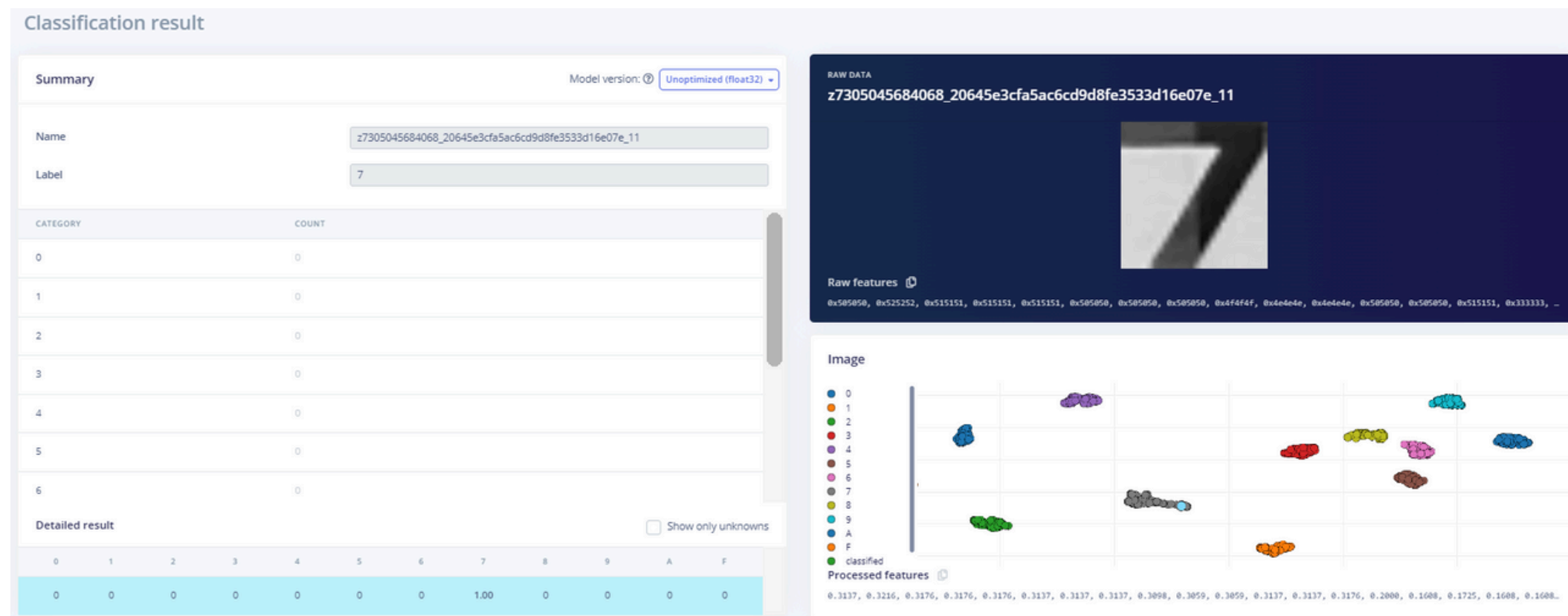


Run this model

Scan QR code or launch in browser to test your prototype

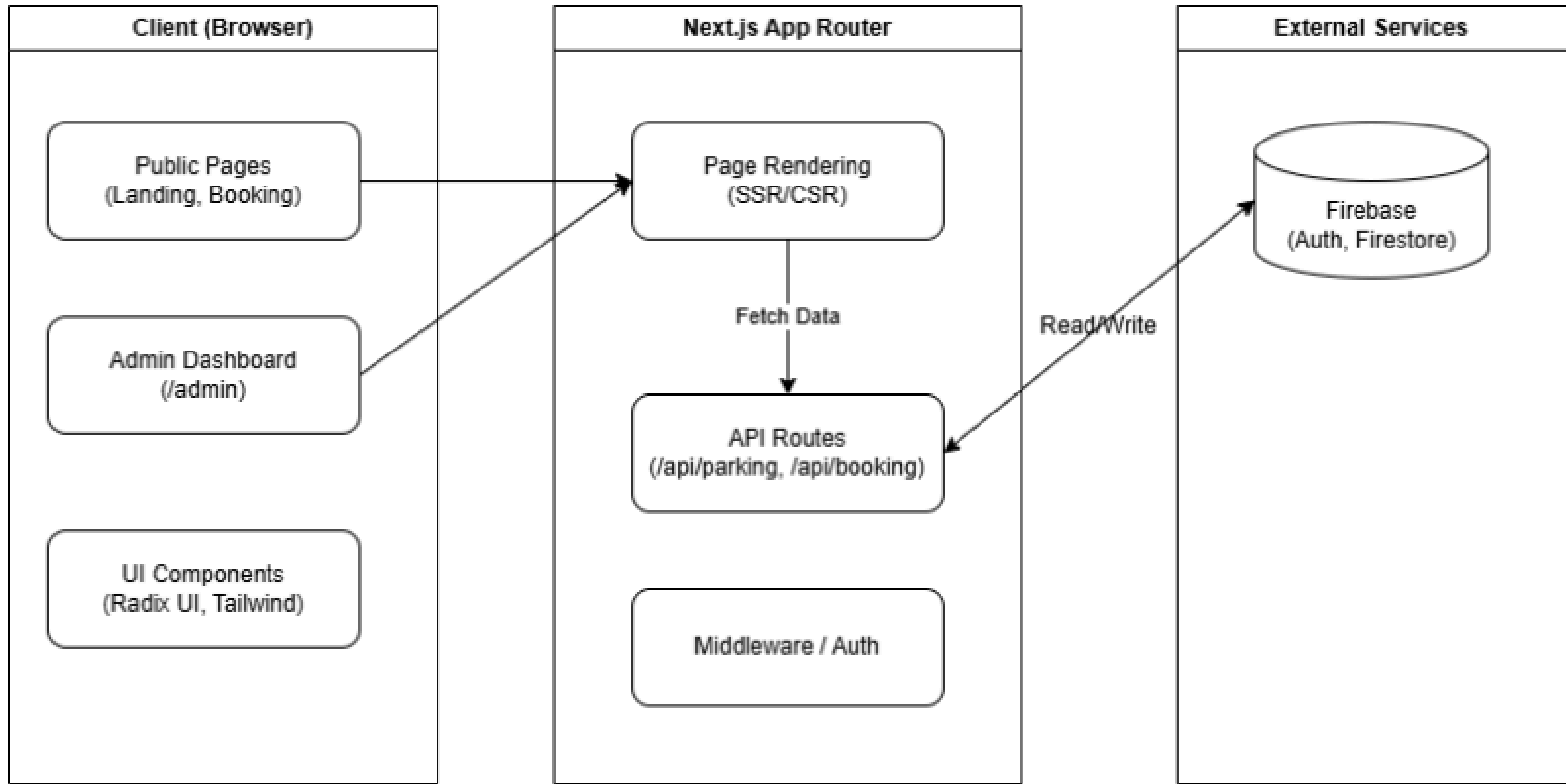


Launch in browser



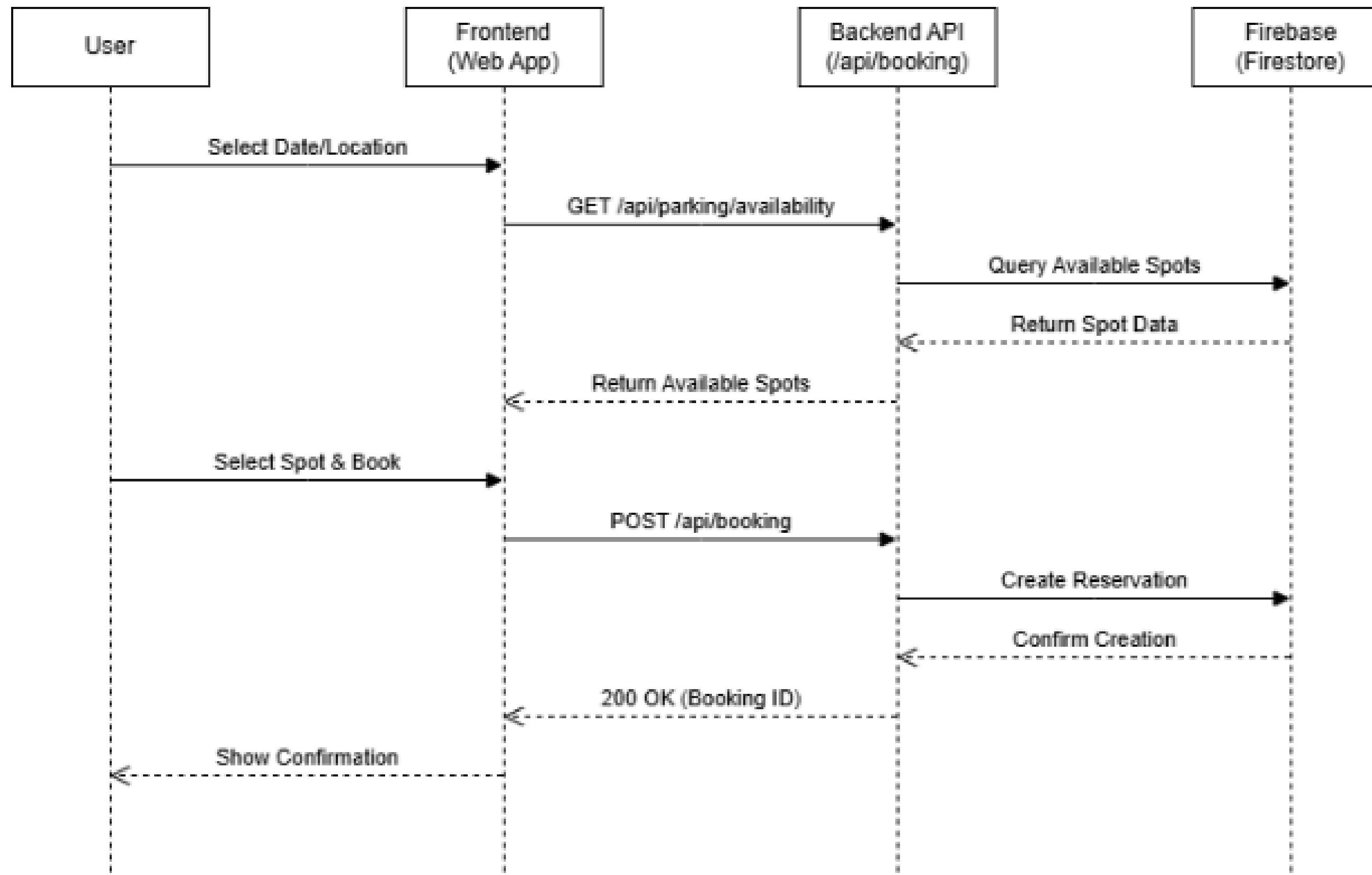
5. WEB & MOBILE

Technology Stack



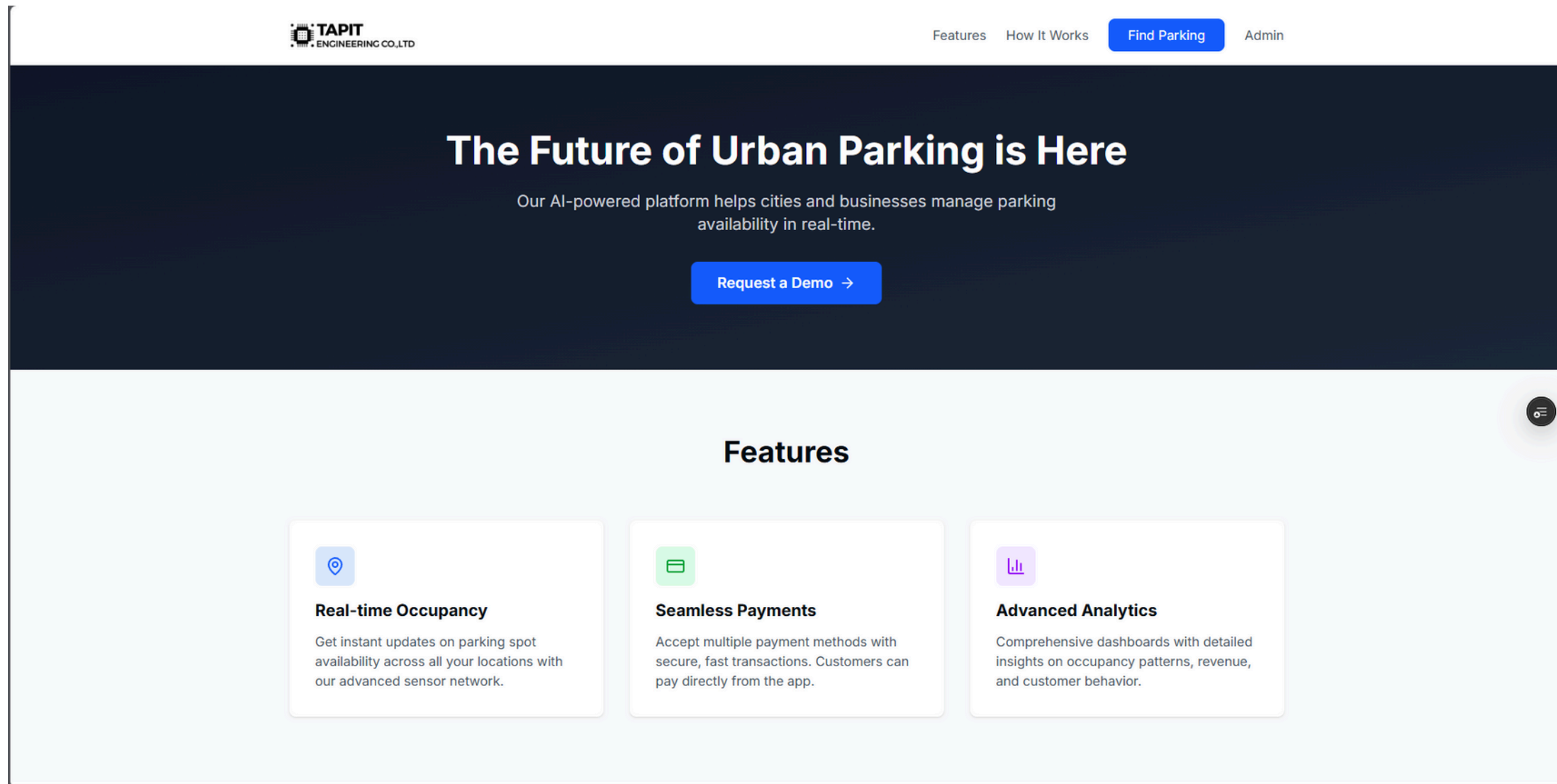
5. WEB & MOBILE

Sequence diagram



5. WEB & MOBILE

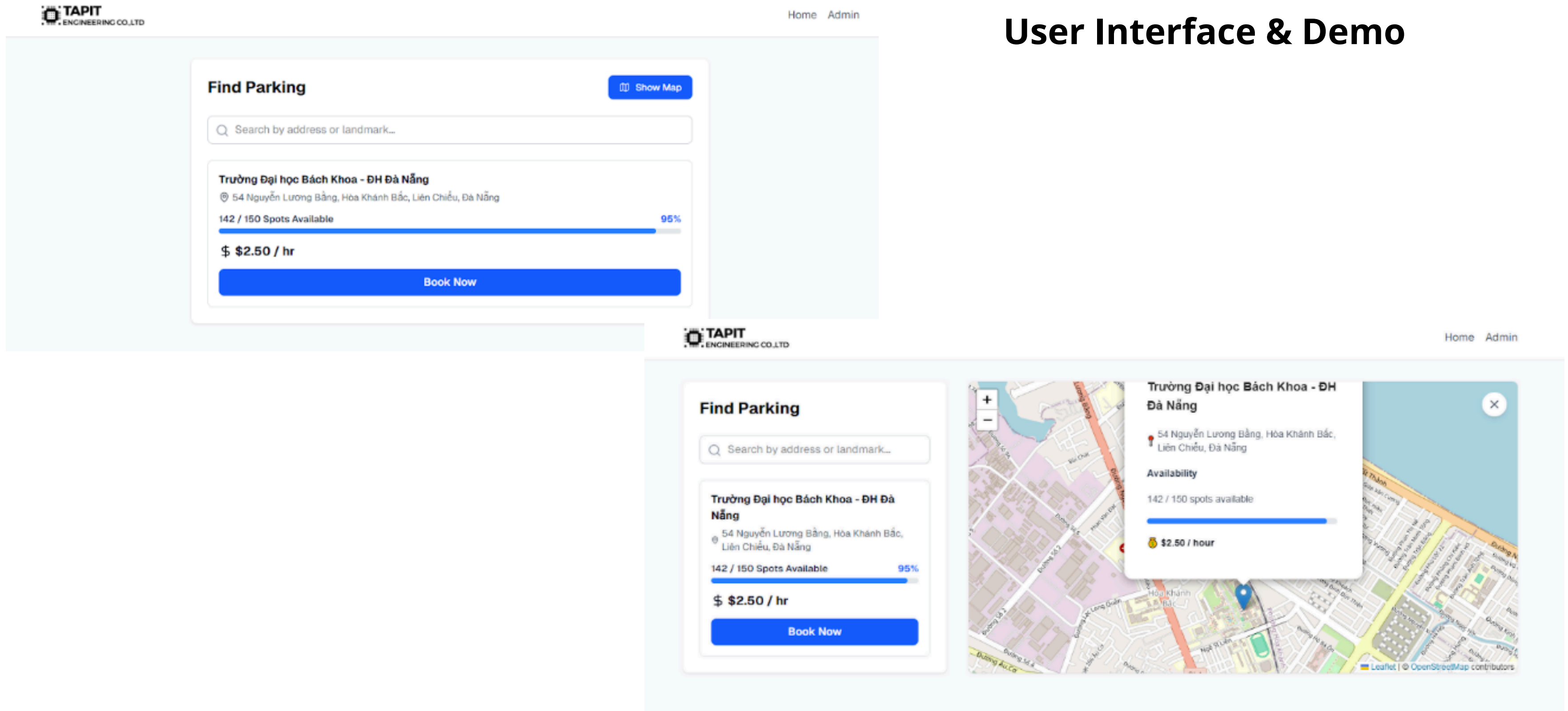
User Interface & Demo



The main home page for the user to book a slot.

5. WEB & MOBILE

User Interface & Demo



Searching for a parking slot.

5. WEB & MOBILE

User Interface & Demo

Book Parking Spot

Trường Đại học Bách Khoa - ĐH Đà Nẵng

Personal Information

Full Name

John Doe

Email

john@example.com

Phone Number

+84 123 456 789

License Plate

ABC-123

Booking Details

Date

24/11/2025

Time

08:39 SA

Duration (hours)

2 hours

Total Cost

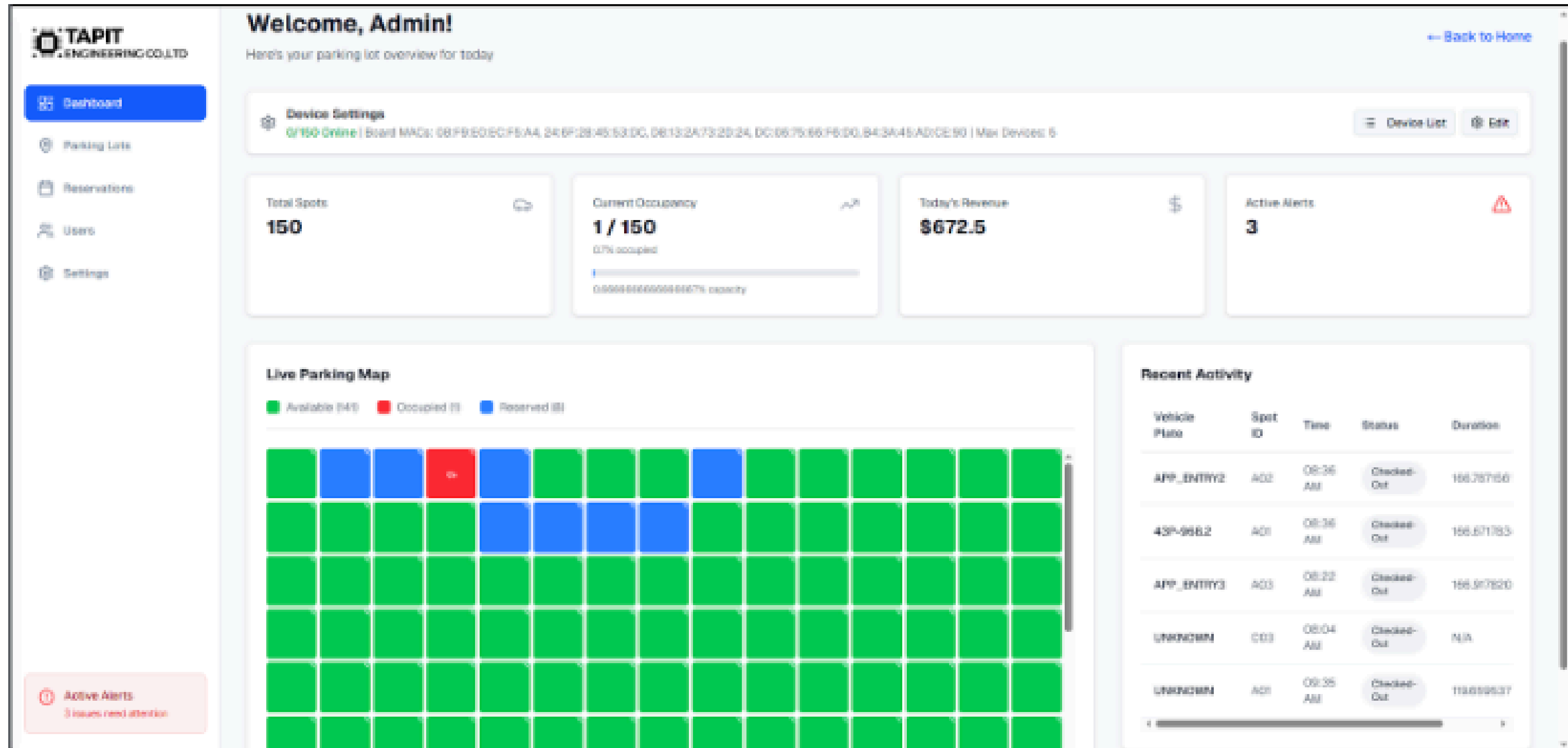
\$5.00

Confirm Booking

Register form for booking users.

5. WEB & MOBILE

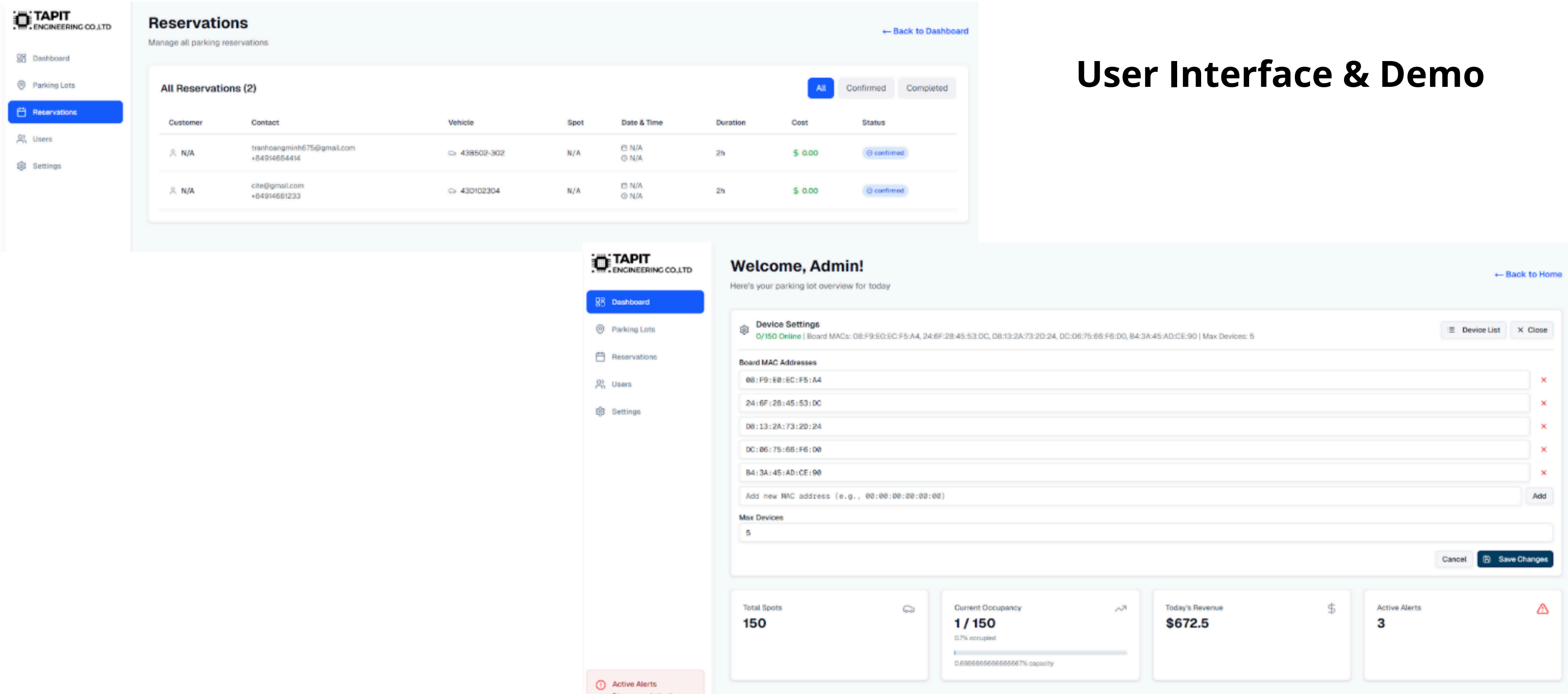
User Interface & Demo



Monitoring dashboard page for admin.

5. WEB & MOBILE

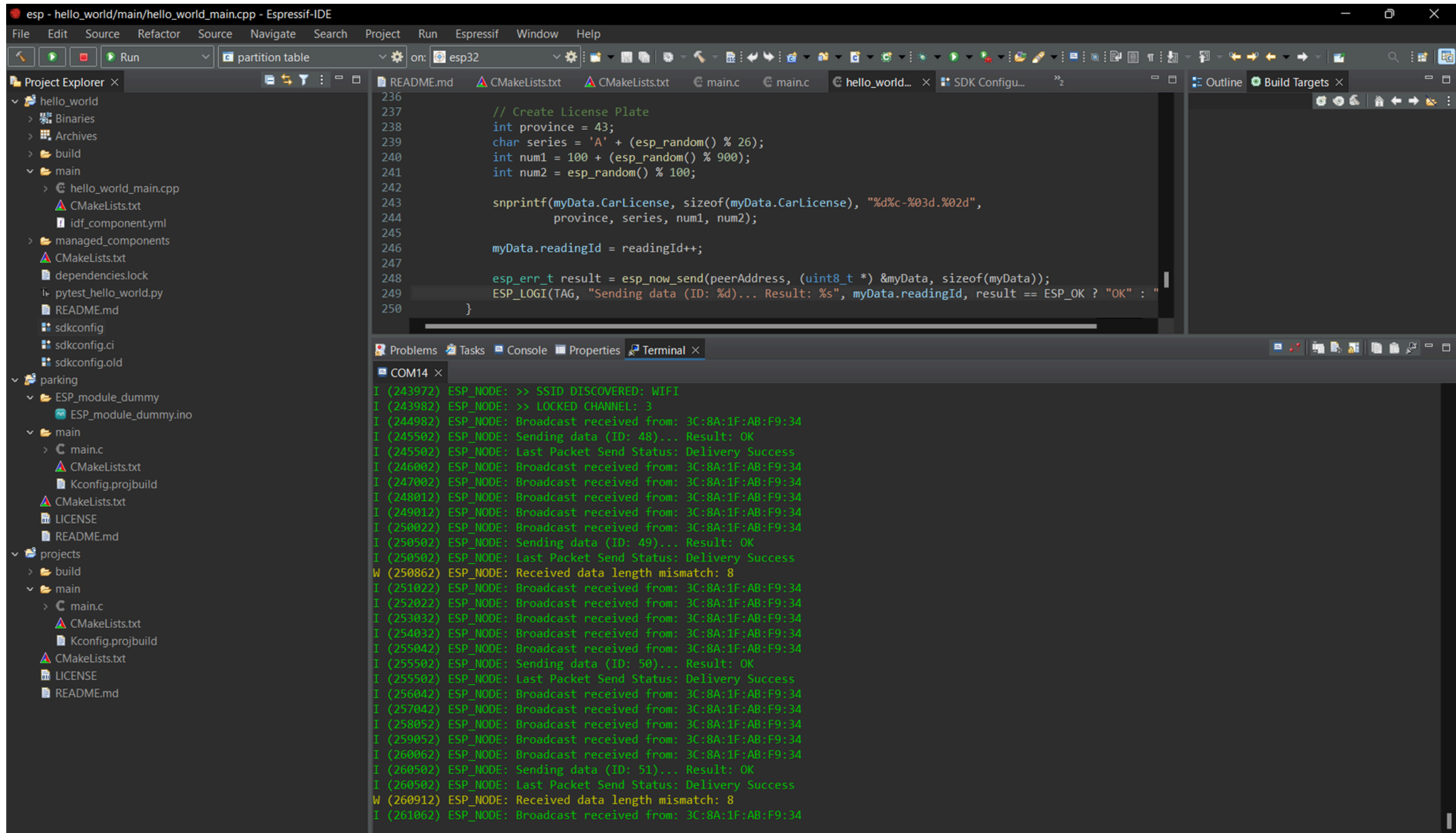
User Interface & Demo



A Reserving booking page. & The administrator page for device configuration.

ESP NOW

6. RESULT

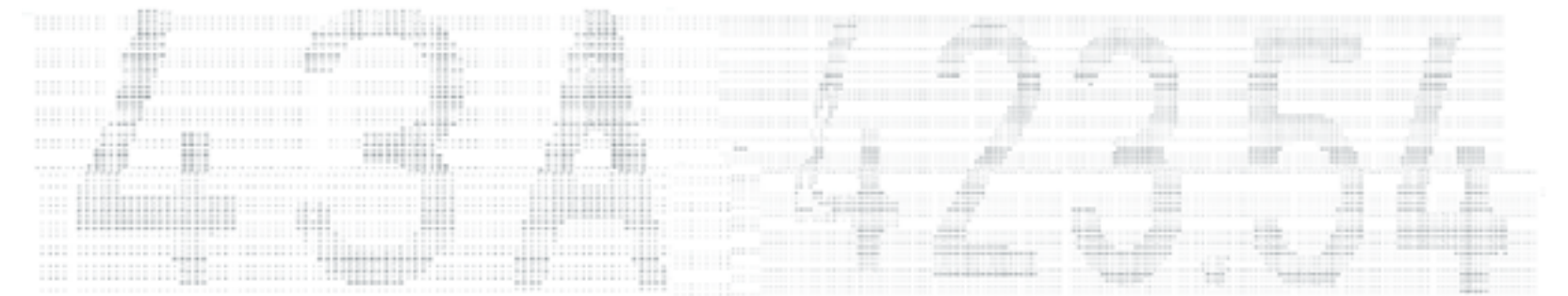
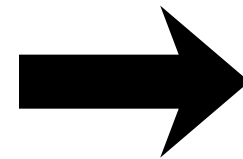


The screenshot displays the Espressif IDE interface. The left sidebar shows the Project Explorer with a tree view of the project files, including 'hello_world' and 'parking'. The main editor window shows the source code for 'hello_world_main.cpp', which includes a license plate generation function and an ESP-NOW sending function. The bottom panel shows the Terminal output, which displays the device's boot logs and the successful transmission of data packets.

```
236
237 // Create License Plate
238 int province = 43;
239 char series = 'A' + (esp_random() % 26);
240 int num1 = 100 + (esp_random() % 900);
241 int num2 = esp_random() % 100;
242
243 snprintf(myData.CarLicense, sizeof(myData.CarLicense), "%d%c-%03d.%02d",
244          province, series, num1, num2);
245
246 myData.readingId = readingId++;
247
248 esp_err_t result = esp_now_send(peerAddress, (uint8_t *) &myData, sizeof(myData));
249 ESP_LOGI(TAG, "Sending data (ID: %d)... Result: %s", myData.readingId, result == ESP_OK ? "OK" : "Error");
250 }
```

```
I (243972) ESP_NODE: >> SSID DISCOVERED: WIFI
I (243982) ESP_NODE: >> LOCKED CHANNEL: 3
I (244982) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (245502) ESP_NODE: Sending data (ID: 48)... Result: OK
I (245502) ESP_NODE: Last Packet Send Status: Delivery Success
I (246002) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (247002) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (248012) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (249012) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (250022) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (250502) ESP_NODE: Sending data (ID: 49)... Result: OK
I (250502) ESP_NODE: Last Packet Send Status: Delivery Success
W (250862) ESP_NODE: Received data length mismatch: 8
I (251022) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (252022) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (253032) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (254032) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (255042) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (255502) ESP_NODE: Sending data (ID: 50)... Result: OK
I (255502) ESP_NODE: Last Packet Send Status: Delivery Success
I (256042) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (257042) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (258052) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (259052) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (260062) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
I (260502) ESP_NODE: Sending data (ID: 51)... Result: OK
I (260502) ESP_NODE: Last Packet Send Status: Delivery Success
W (260912) ESP_NODE: Received data length mismatch: 8
I (261062) ESP_NODE: Broadcast received from: 3C:8A:1F:AB:F9:34
```

RESULT AT EDGE



Capture an image with the view in front of a car with a size of 640*480.

Using the DSP algorithm to find and crop the licence plate number region.



Xử lý ký tự 1: X[31-63] (Kích thước: 33x68)

✓ Nhận dạng: 4 (độ tin cậy: 0.996)

Xử lý ký tự 2: X[75-108] (Kích thước: 34x68)

✓ Nhận dạng: 3 (độ tin cậy: 0.996)

Xử lý ký tự 3: X[118-148] (Kích thước: 31x68)

✓ Nhận dạng: A (độ tin cậy: 0.996)

Xử lý ký tự 4: X[183-215] (Kích thước: 33x68)

✓ Nhận dạng: 4 (độ tin cậy: 0.996)

Xử lý ký tự 5: X[225-258] (Kích thước: 34x69)

✓ Nhận dạng: 2 (độ tin cậy: 0.996)

Xử lý ký tự 6: X[268-302] (Kích thước: 35x69)

✓ Nhận dạng: 3 (độ tin cậy: 0.996)

Xử lý ký tự 7: X[324-358] (Kích thước: 35x69)

✓ Nhận dạng: 5 (độ tin cậy: 0.996)

Xử lý ký tự 8: X[367-395] (Kích thước: 29x70)

✓ Nhận dạng: 4 (độ tin cậy: 0.938)

🎯 === KẾT QUẢ NHẬN DẠNG: 43A42354 ===

Step 3: Using the DSP algorithm to crop and resize to 28*28 for each letter.

Step 4: Load the AI model and normalise the letter to a suitable form of AI model input, and use the AI to find the license plate number.

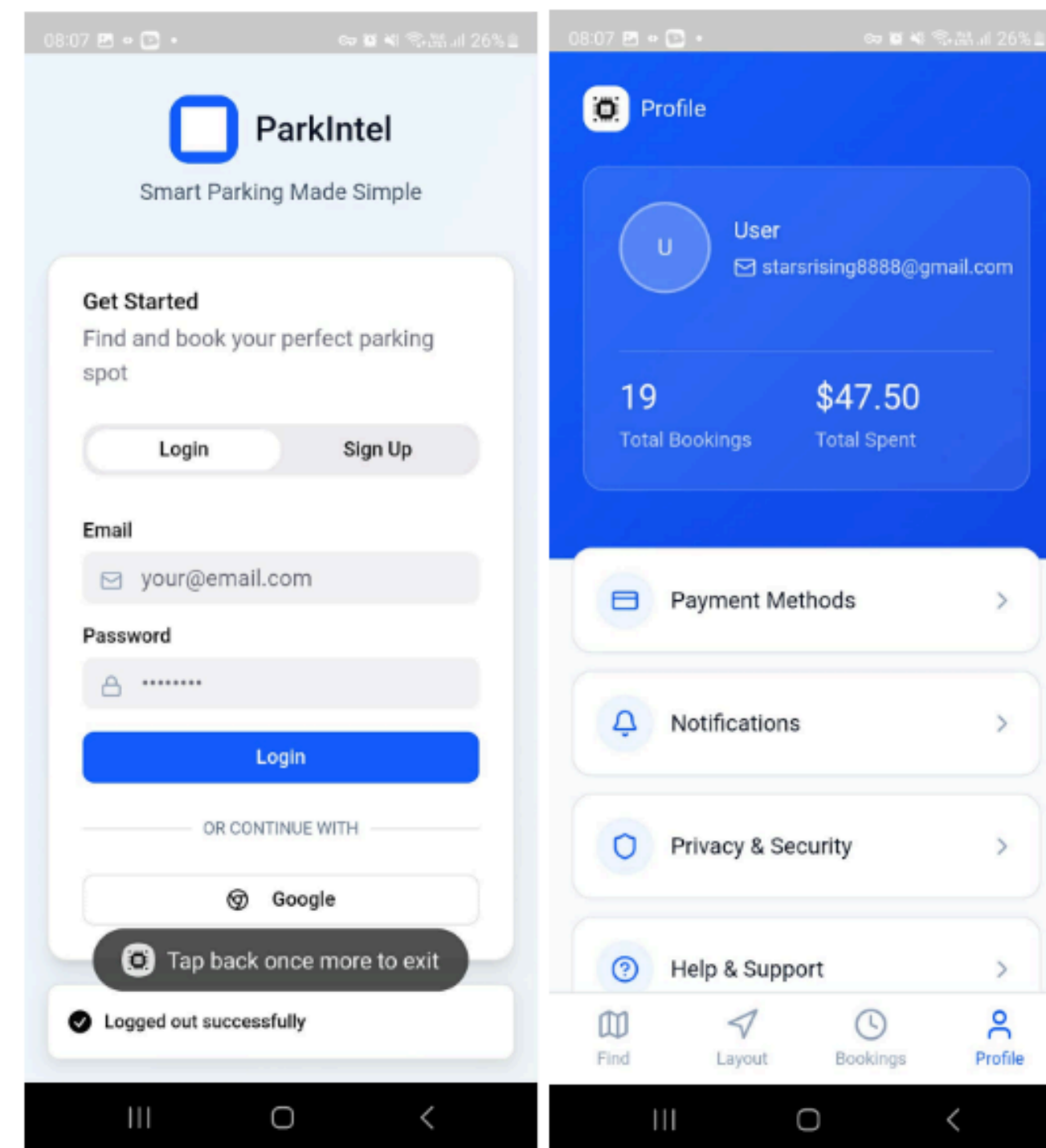
6. RESULT

Experimental Results

Connected Devices (0/150 Online)
List of all parking spot devices and their connection status.

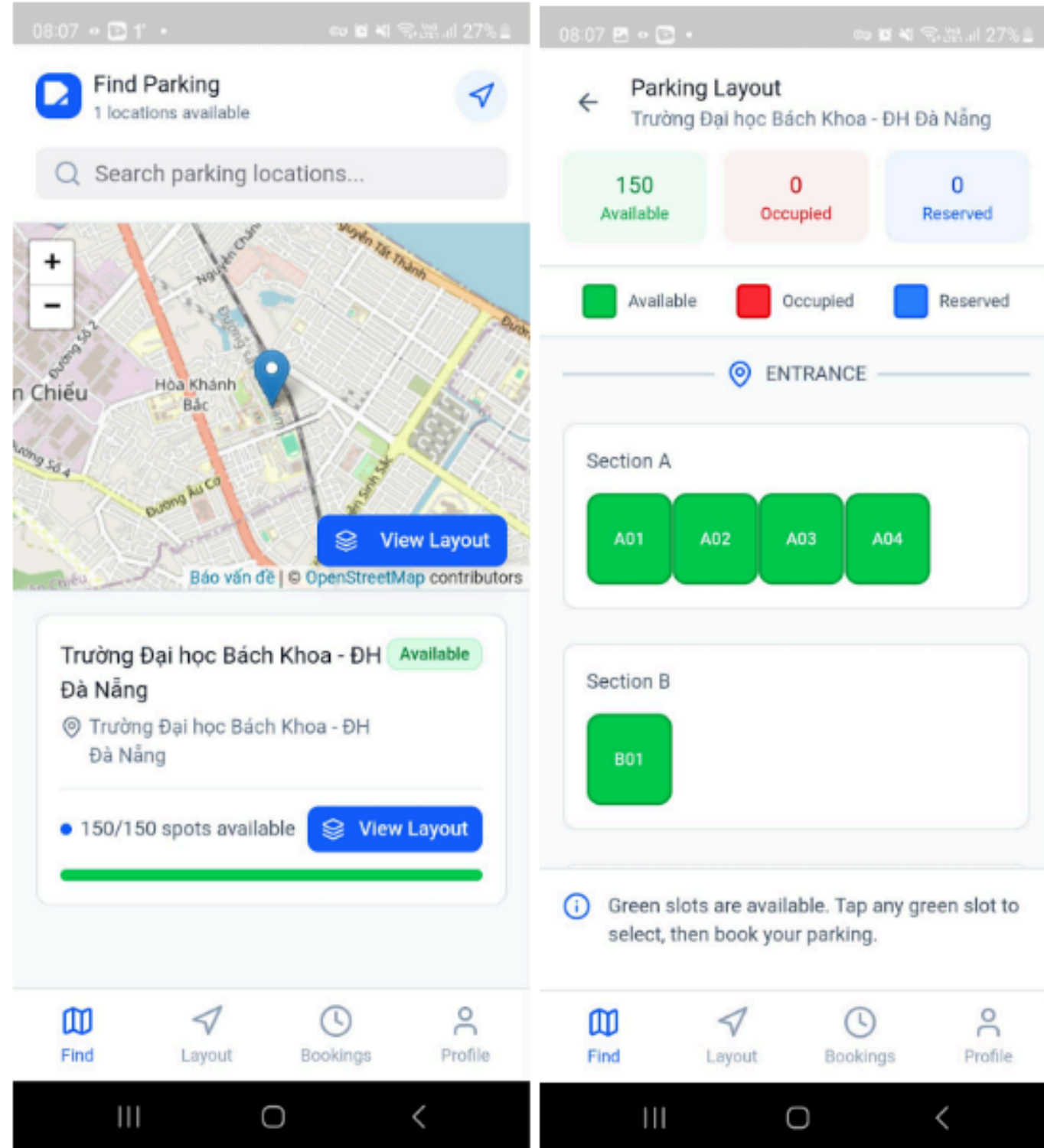
Spot ID	Status	MAC Address	Action
A01	offline	N/A	
A02	offline	N/A	
A03	offline	N/A	
A04	offline	N/A	
A05	offline	N/A	
B01	offline	N/A	
C01	offline	N/A	
C02	offline	N/A	
C03	offline	N/A	
C04	offline	N/A	
C05	offline	N/A	
C06	offline	N/A	

App display screen of connected devices for users.

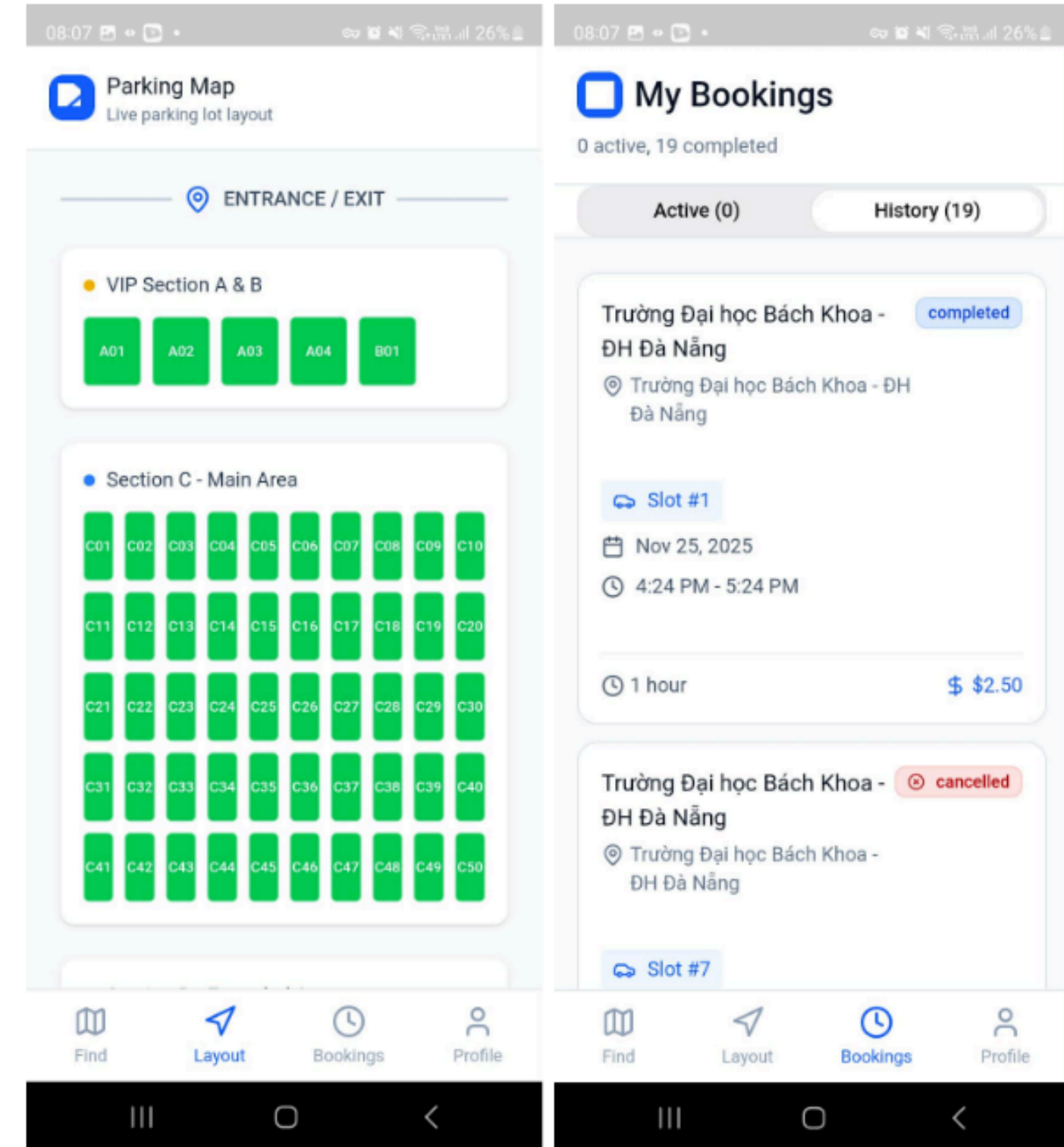


Main page and functional pages.

6. RESULT



Register form for booking
users in user app



Monitoring dashboard page

7. CONCLUSION

Key Achievements

- System Implementation: Successfully developed the IoT-based Smart Parking System using a Star Topology network.
- Real-time Monitoring: Achieved low-latency status updates to the Web Interface and Mobile App by integrating sensors and cameras.
- Efficient Communication: Implemented the ESP-NOW protocol, ensuring energy efficiency and stable connectivity between Sensor Nodes and the Gateway.
- Centralized Management: Established accurate real-time data synchronization with Firebase for effective administration.

Limitations

- Image Processing: License plate recognition accuracy is currently sensitive to environmental lighting conditions.
- Infrastructure Dependency: The Gateway requires a stable Internet connection to synchronize data with the Cloud.

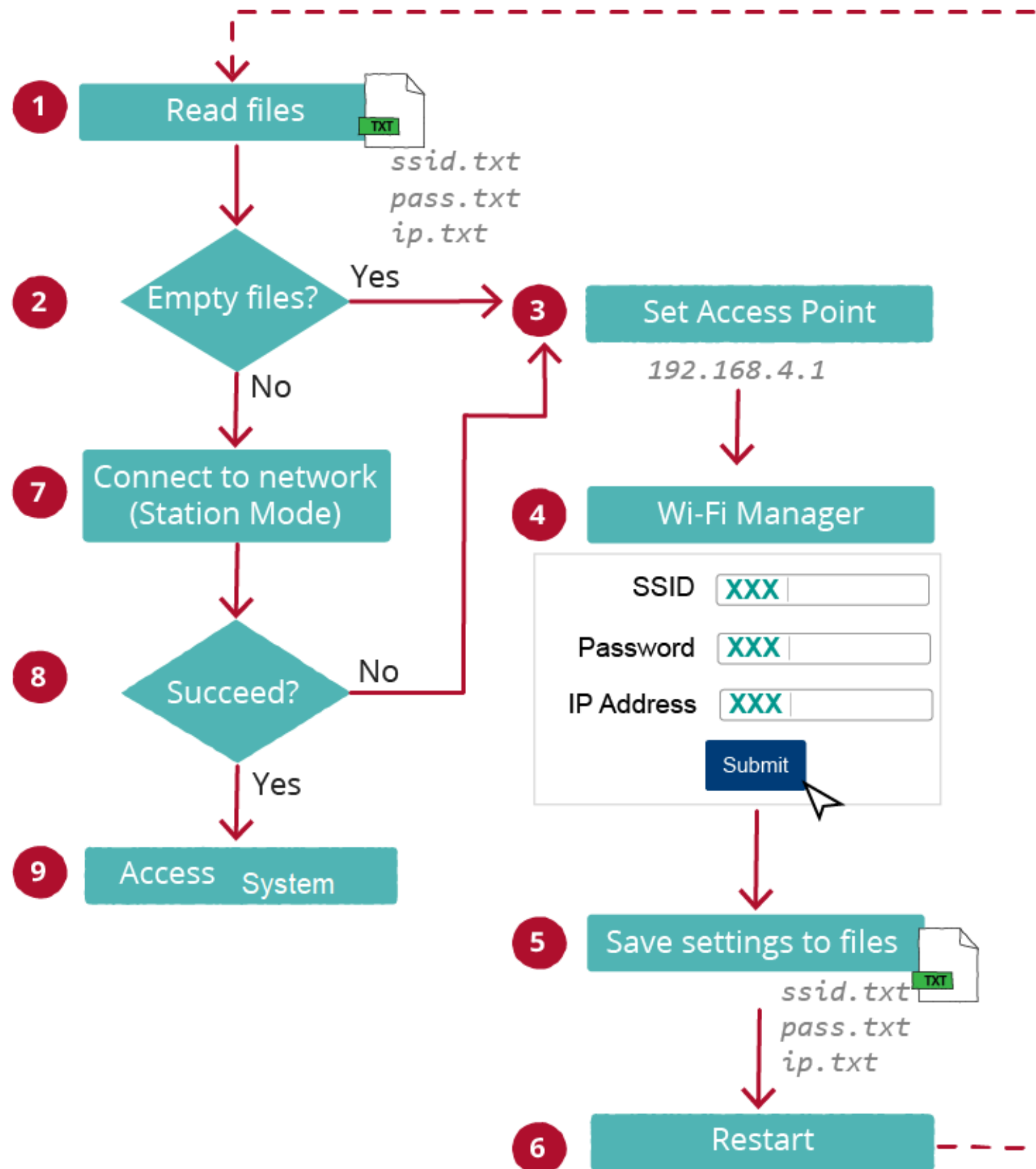
7. CONCLUSION

Future Work

- AI Integration: Implement Deep Learning algorithms to improve license plate recognition accuracy in complex environments.
- E-Payment System: Add automatic payment features via the Mobile App (e.g., QR Code or E-wallets).
- Green Energy: Explore solar power solutions for outdoor independent sensor nodes.

The background features several blue geometric shapes, including triangles and parallelograms, positioned at the corners and edges of the slide. Some shapes have white outlines.

THANK YOU



- When the ESP main dashboard first starts, it tries to read the ssid.txt, pass.txt and ip.txt files* (1);
- If the files are empty (2) (the first time you run the board, the files are empty), your board is set as an access point (3);
- Using any Wi-Fi enabled device with a browser, you can connect to the newly created Access Point (default name ESP-WIFI-MANAGER);
- After establishing a connection with the ESP-WIFI-MANAGER, you can go to the default IP address 192.168.4.1 to open a web page that allows you to configure your SSID and password (4);
- The SSID, password, and IP address inserted in the form are saved in the corresponding files: ssid.txt, pass.txt, and ip.txt (5);
- After that, the ESP board restarts (6);
- This time, after restarting, the files are not empty, so the ESP will try to connect to the network in station mode using the settings you've inserted in the form (7);
- If it establishes a connection, the process is completed successfully, and you can access the main web server page that can do whatever you want (control sensor readings, control outputs, display some text, etc.) (9). Otherwise, it will set the Access Point (3), and you can access the default IP address (192.168.4.1) to add another SSID/password combination.